

Inverse-CDF sampling of a power-law prior on $\log P$

How `sample_logP_powerlaw` turns a uniform draw u into a sample $x = \log_{10}(P/\text{day})$

The function draws values of $x = \log_{10}(P/\text{day})$ from the prior

$$p(x) \propto x^\pi \quad \text{on} \quad [a, b] = [\log P_{\min}, \log P_{\max}], \quad a, b > 0.$$

This is the period-distribution prior used by Dsilva+ (2023) for WR+OB systems. The exponent π is the Validation-tab slider value: it sets how steeply the prior favours short or long log-periods.

Because $p(x)$ has a closed-form integral, the function uses INVERSE-CDF (inverse-transform) sampling: it draws $u \sim \text{Uniform}(0, 1)$, then returns $x = F^{-1}(u)$ where F is the CDF of p . This is exact — no rejection sampling, no importance weights, no truncation bias.

There is one degenerate case: when $\pi = -1$ the integral of x^π becomes a logarithm rather than a power, so a separate closed-form is used. Everything else follows the same recipe.

What follows: derivation (page 2), prior shapes (page 3), the inverse-CDF in pictures (page 4), validation against the analytic PDF (page 5), and the corresponding distribution of orbital periods in days (page 6).

Derivation: where $\pi + 1$ comes from

1. Start from the unnormalised PDF

$$p(x) \propto x^\pi \text{ on } [a, b], \quad a, b > 0.$$

2. Integrate to get the CDF (this is where the $+1$ appears)

$$\int x^\pi dx = \frac{x^{\pi+1}}{\pi+1} + C \quad (\text{provided } \pi \neq -1)$$

$$\text{so } F(x) = \frac{x^{\pi+1} - a^{\pi+1}}{b^{\pi+1} - a^{\pi+1}}.$$

3. Invert: solve $F(x) = u$ for x

$$x^{\pi+1} = u(b^{\pi+1} - a^{\pi+1}) + a^{\pi+1}$$

$$x = [u(b^{\pi+1} - a^{\pi+1}) + a^{\pi+1}]^{1/(\pi+1)}$$

4. The $\pi = -1$ singular case

$$\int x^{-1} dx = \ln x, \quad \text{so } F(x) = \frac{\ln(x/a)}{\ln(b/a)}$$
$$\implies x = a(b/a)^u.$$

This branch avoids the $1/(\pi+1) = 1/0$ blow-up in the general formula.

5. What the code does

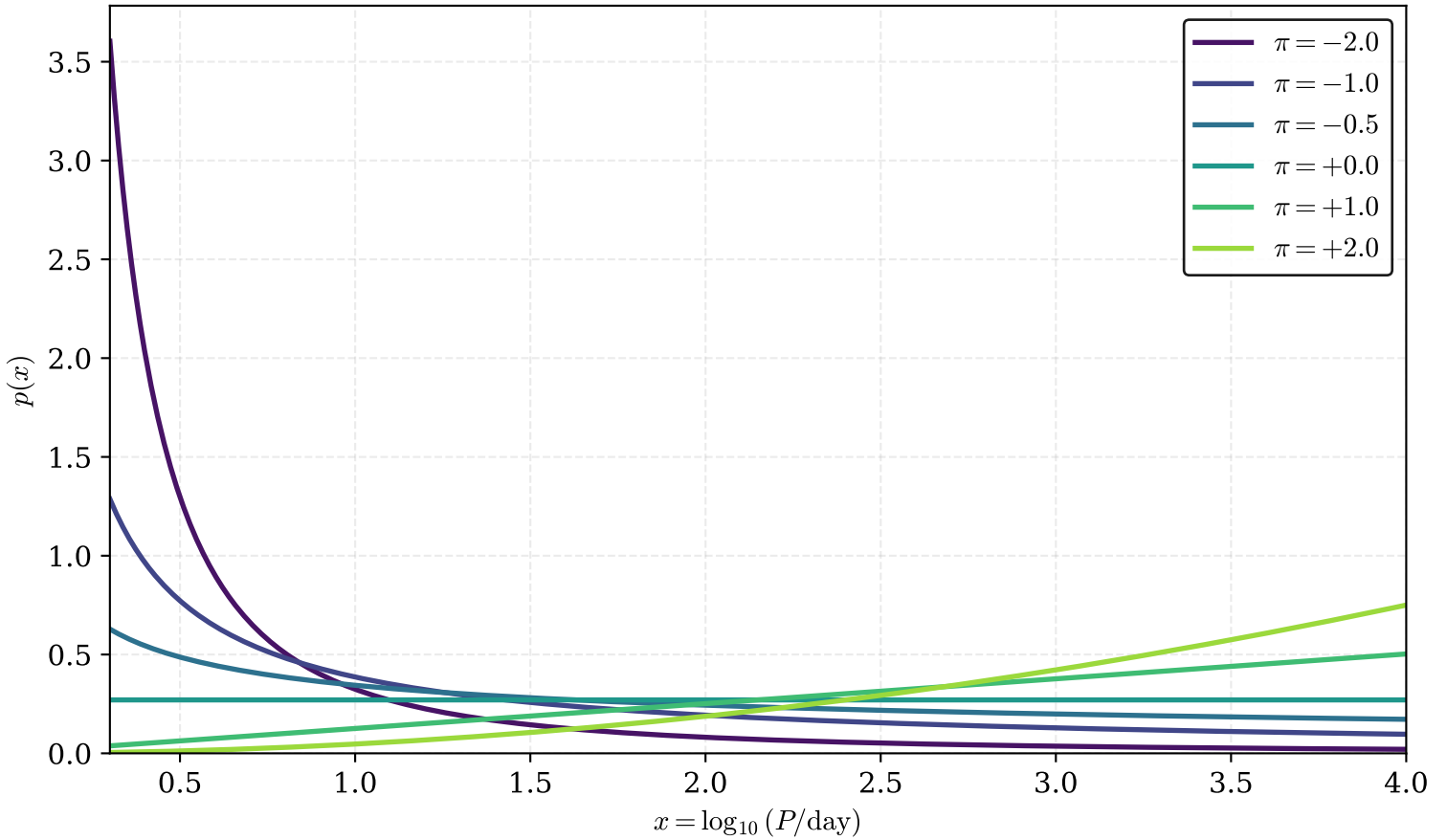
draw $u \sim \text{Uniform}(0, 1)$ (line: `u = rng.random(size)`)

if $|\pi + 1| < 10^{-8}$: $x = a(b/a)^u$ (logarithmic branch)

else: set `exponent` = $\pi + 1$, then $x = (u(b^{\pi+1} - a^{\pi+1}) + a^{\pi+1})^{1/(\pi+1)}$

Mnemonic: integrating x^π lifts the exponent by 1, and inverting the CDF takes the $1/(\pi+1)$ root that undoes it

Power-law prior $p(x) \propto x^\pi$ on $[a, b] = [0.3, 4.0]$ (normalised)

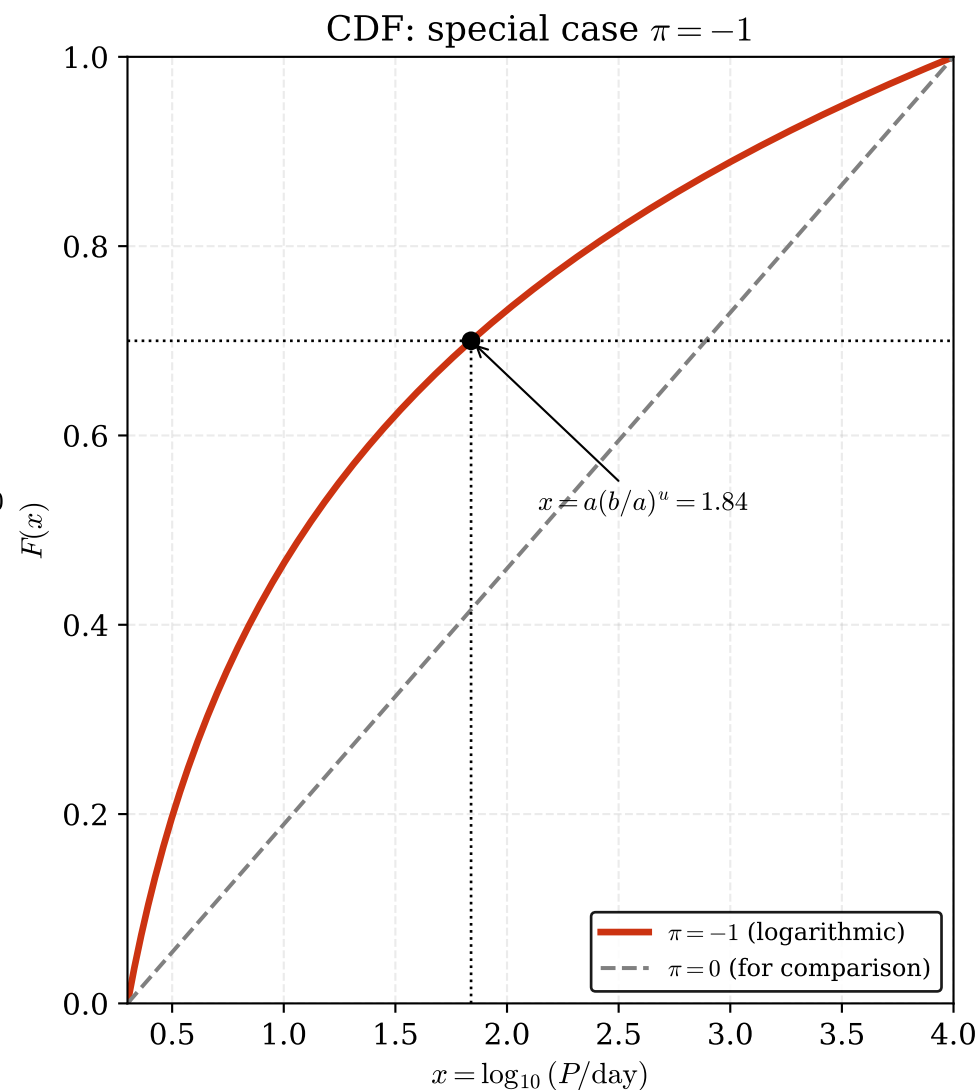
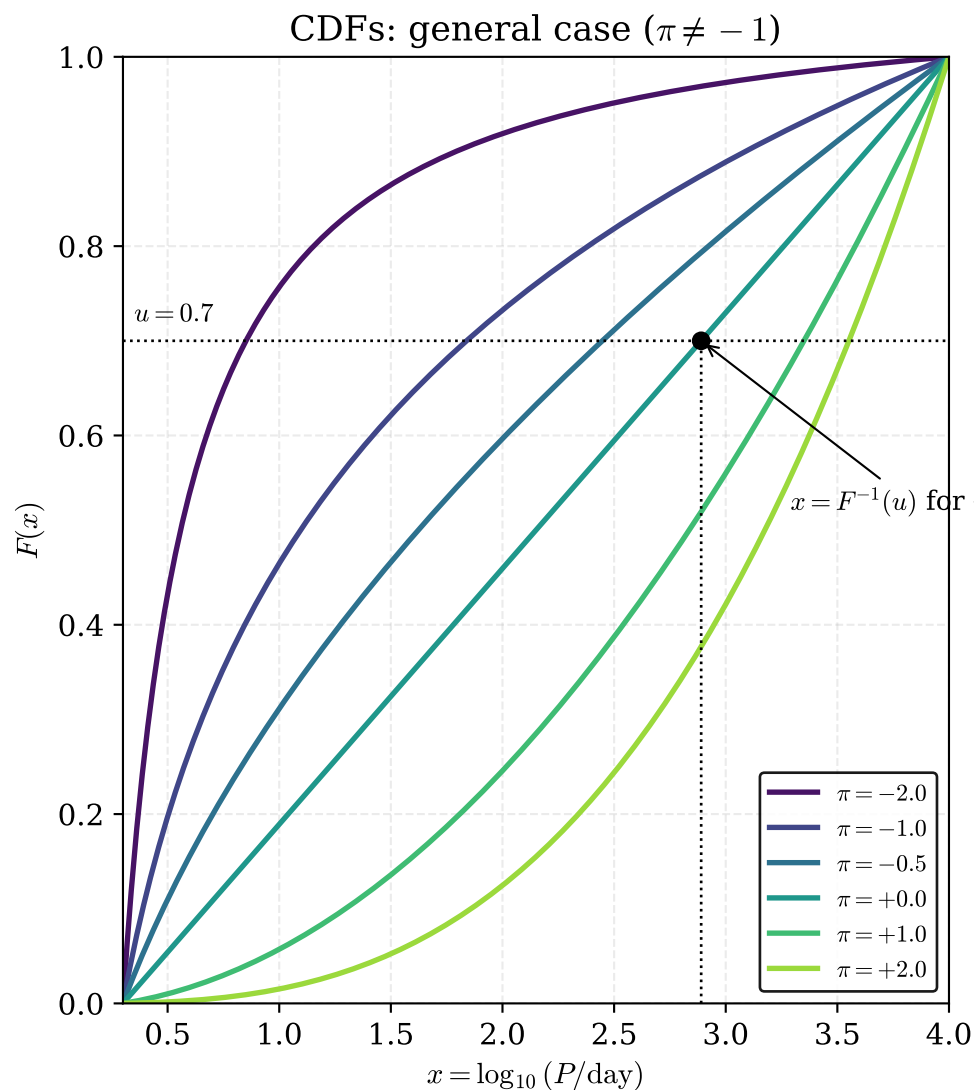


How to read this:

- Each curve is the prior on $x = \log_{10} P$, normalised to integrate to 1 over the x -range.
- $\pi = 0$ (flat) is Öpik's law — uniform in $\log P$. This is the textbook 'no-information' prior.
- $\pi > 0$ pushes probability towards larger $\log P$ (long-period systems).
- $\pi < 0$ pushes probability towards small $\log P$ (short-period, RV-active systems).
- $\pi = -1$ is the boundary case: $p(x) \propto 1/x$, equivalent to a flat distribution in $\ln(\log P)$ — scale-invariant on the log-period axis.

Recovery exercise: given a measured ΔRV distribution from real WR stars, the bias-correction grid asks 'which π best reproduces it?'. The Validation tab tests that recovery by injecting a known π and checking it back out.

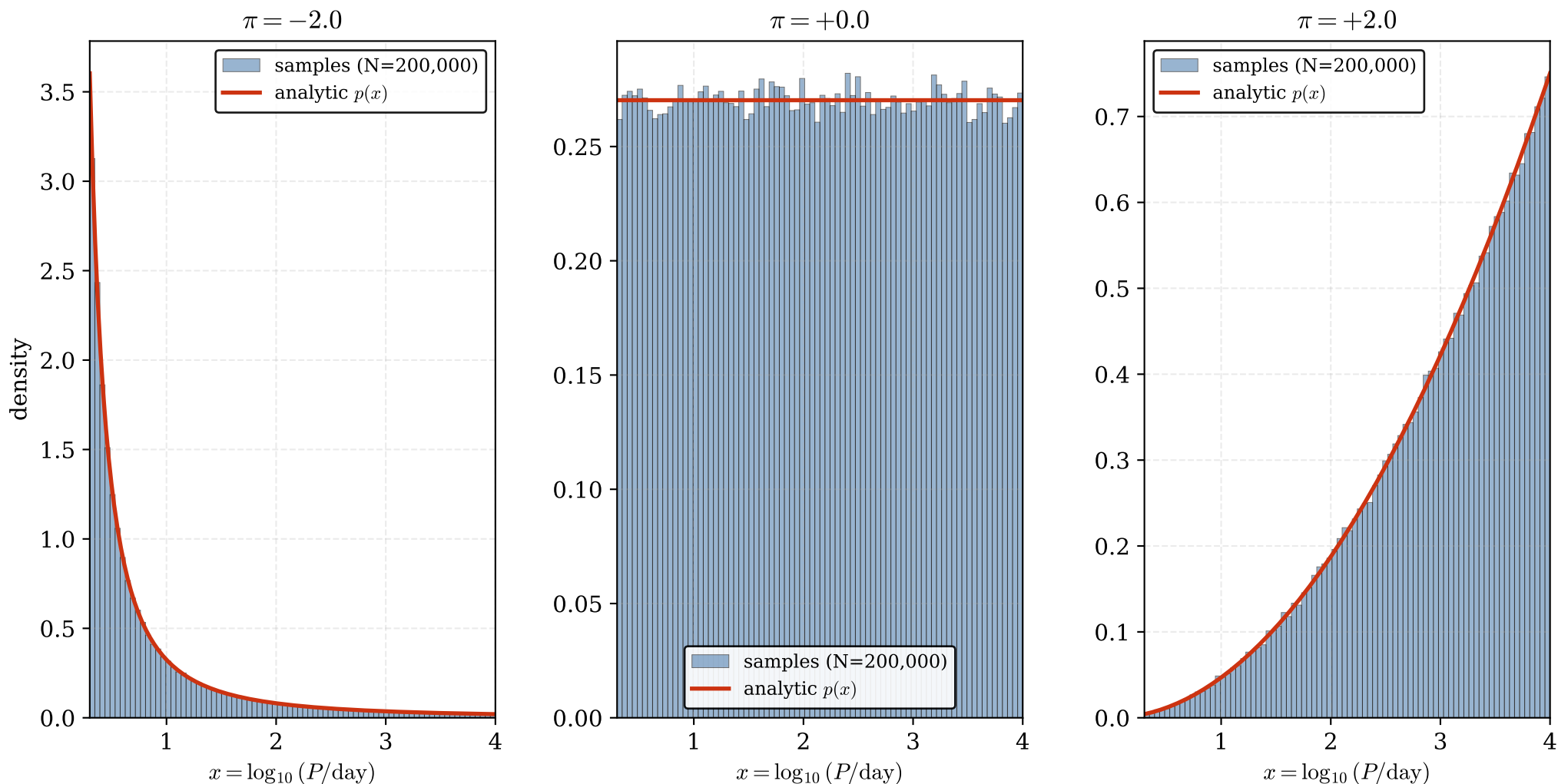
Inverse-CDF sampling: pick u on the y -axis, drop down to read off x



Reading the figure

For each $u \in [0, 1]$ the CDF curve gives a unique x . Drawing u uniformly therefore yields x distributed as $p(x)$. The dotted construction shows one example with $u = 0.7$. The right panel uses the logarithmic CDF that the code falls back to when $\pi = -1$ (because the general formula divides by zero).

Validation: histogram of sample_logP_powerlaw output vs analytic $p(x)$

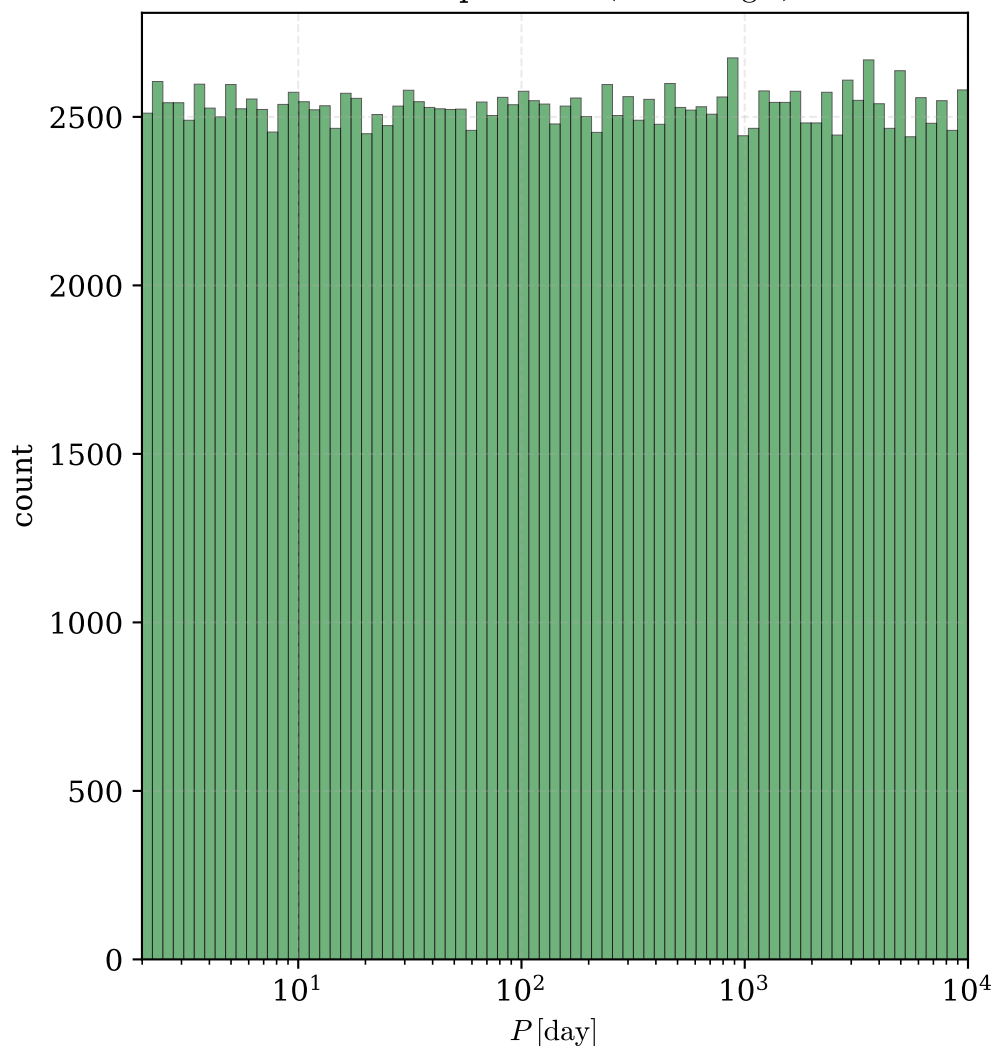


What this proves

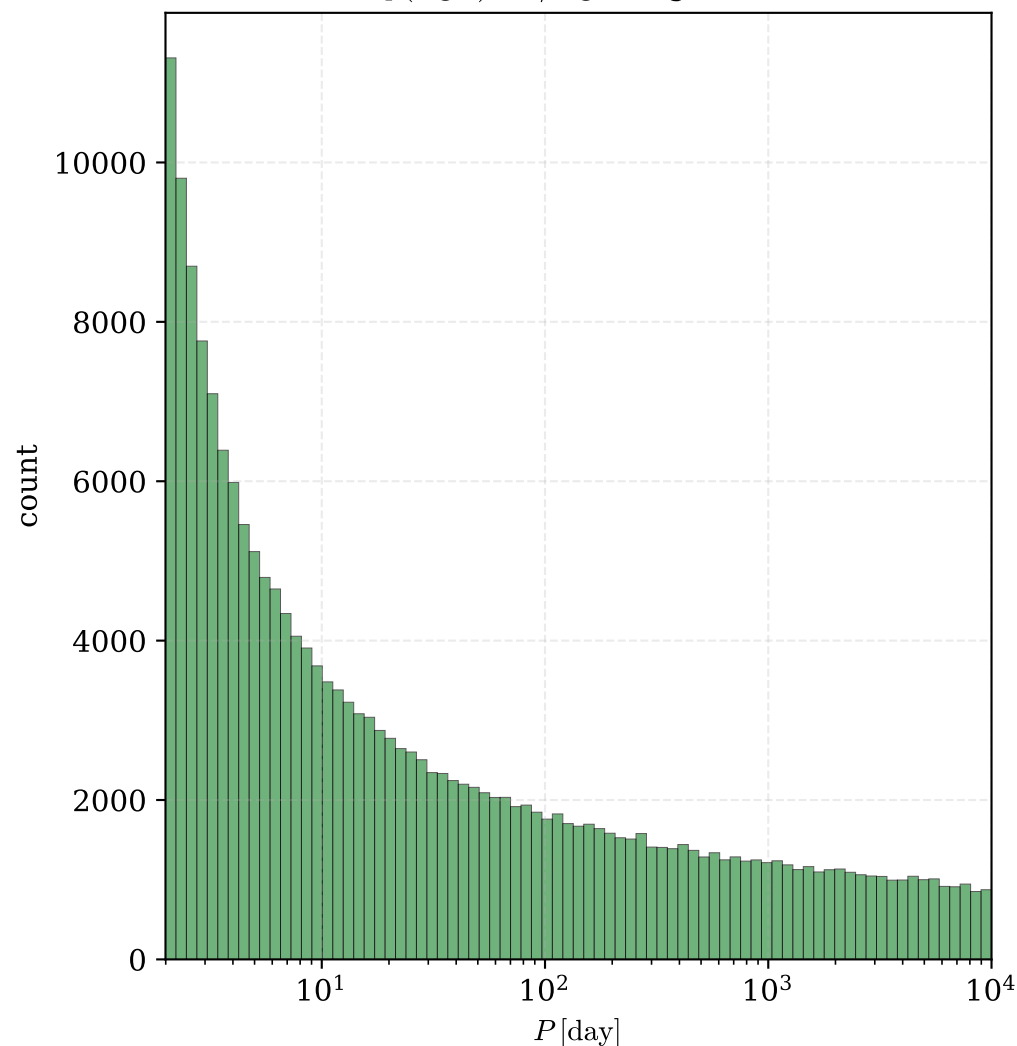
The blue histograms are $N=200,000$ samples drawn by the production `sample_logP_powerlaw` function (imported from `wr_bias_simulation.py`). The red curves are the analytic normalised PDFs computed independently. They overlap to within Monte-Carlo noise across all three π values, confirming the inverse-CDF implementation matches the intended distribution. This is the same generator that feeds the mock-data pipeline in the Validation tab: every binary in the synthetic population gets its $\log P$ drawn from exactly this curve.

Same draws, viewed as orbital periods in days

$\pi = 0$ — Öpik's law (flat in $\log P$)



$\pi = -1$ — $p(\log P) \propto 1/\log P$ (logarithmic CDF)



Why two columns?

Both panels show the same draws as page 5, transformed by $P = 10^x$ and binned in $\log P$. Left: $\pi = 0$ — flat in $\log P$

(Öpik's law); the $\log P$ histogram is therefore flat. On a linear P axis it would be strongly skewed to short periods. Right:

$\pi = -1$ — $p(\log P) \propto 1/\log P$, so the $\log P$ histogram rises towards small $\log P$ (i.e. short orbital periods). It

is NOT uniform in P : that would require $p(\log P) \propto P \ln 10$, an exponential in $\log P$, not a power. Choosing π

in the Validation tab therefore controls how many short-period (RV-active) vs long-period (RV-quiet) binaries the synthetic population contains, which in turn sets the recoverable ΔRV signal.